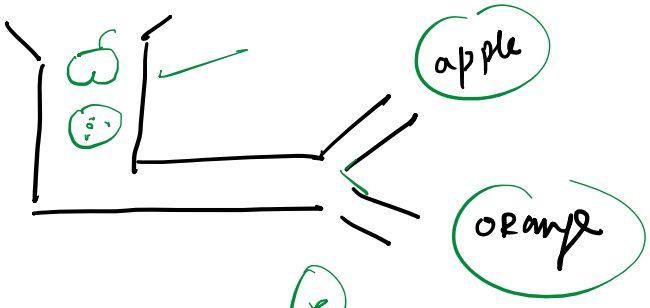
* LINEAR ALGEBRA ⇒ Revision + PROBLEM SOLVING

[Agenda → Revising Previous topics
→ PROBLEM SOLVING ⇒] (*)

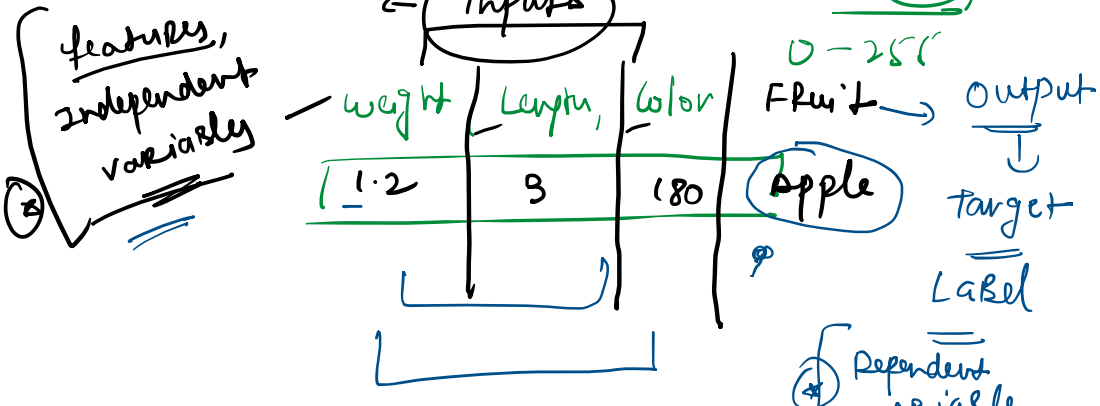
Fruit segregation factory

(*) [Lot → apple & orange]



→ weight, length, (color)*, shape

weight = 1.2, length = 3 cm, color intensity = 180



→ orange / apple

Decision Trees * , LR (Logistic)

Decision Trees * — Regressio — Maths for ML

2 categories → Orange V/S Apple

⇒

2 classes → BINARY Classification

2

more than 2 categories ⇒ (S, M, L, XL)

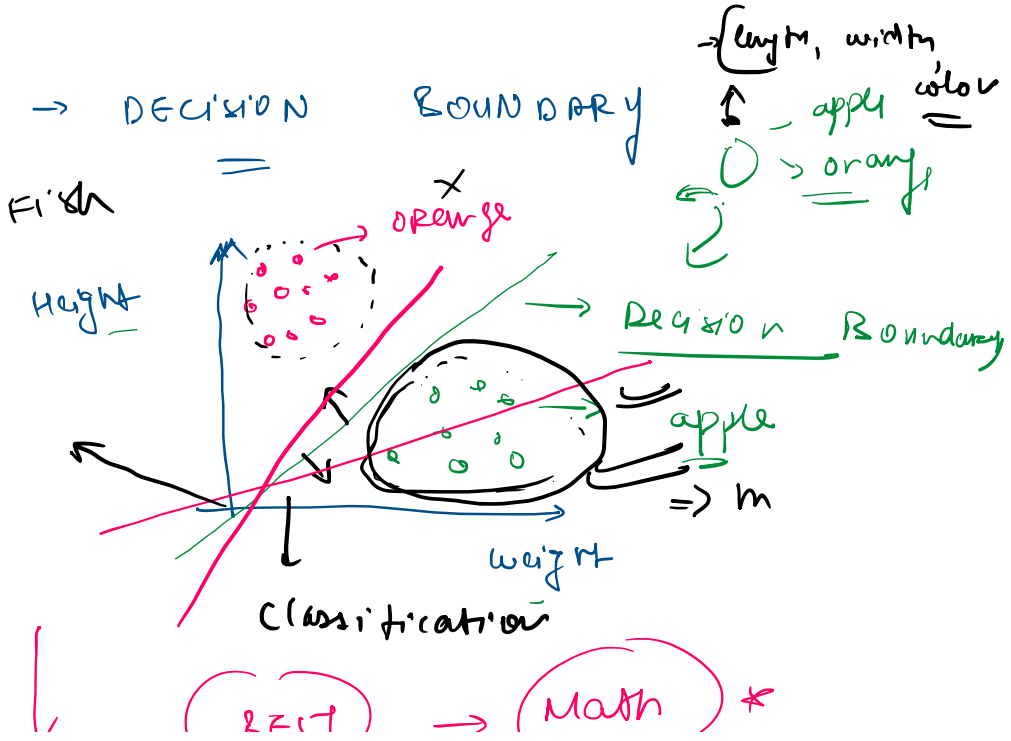
> 2

> 2 ⇒ Multi Class Classification

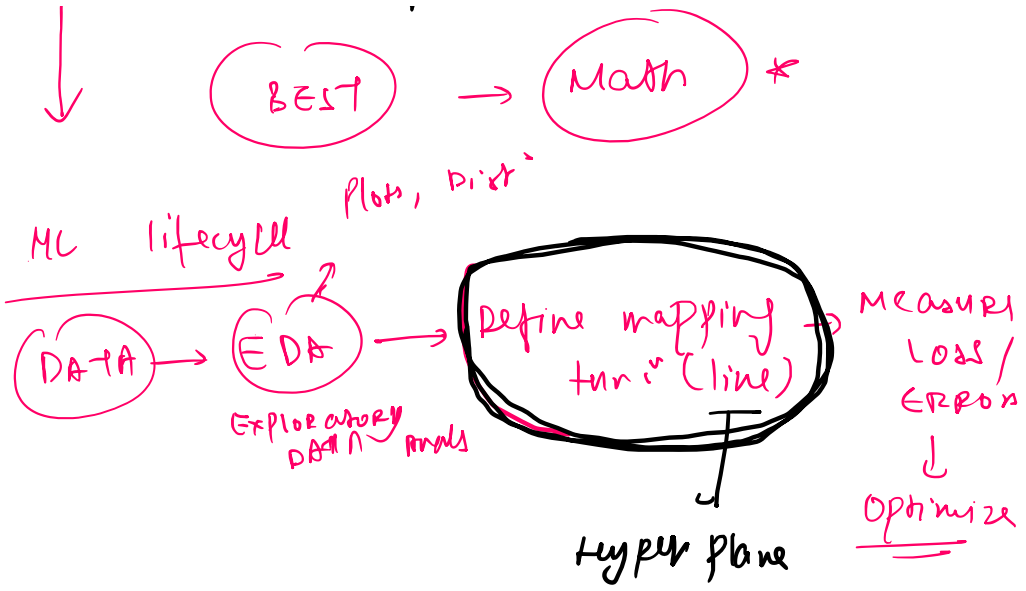
→ Apple, Banana, watermelon, Peach

* MULTI Class ⇒

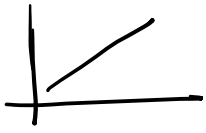
→ DECISION BOUNDARY



Fit → Math *

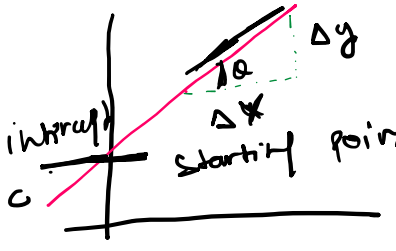


ML → Line ⇒ Hyperplane in 2Dim"



LINES & HYPERPLANES ⇒

⇒ $y = mx + c$ (m: slope, c: intercept)



1 unit → how steep the line

$m = 2$ → every 1 unit ↑ → 2 unit ↑ in y

$m = \tan\theta$

$m = \frac{1}{\sqrt{3}}$, $c = 2$ ⇒ angle line makes with x axis? =

$m = \frac{1}{\sqrt{3}}$, $c = 2$ — with x axis?

$m = \tan\theta = \frac{1}{\sqrt{3}}$, $\theta = 30^\circ$

does this affect angle?

up or down

$y = mx + c$

X

$\theta = 90^\circ$

$\tan\theta$ = undefined

slope intercept f cannot represent vertical line

ML → VERTICAL, HORIZONTAL

POWERFUL FORM

General FORM ⇒

SCHOOL form

$Ax + By + C = 0$

ML form

$w_1 x_1 + w_2 x_2 + w_0 = 0$

w_1: weight; x_1: feature iv/ip; w_0: Bias/intercept

Learning in ML ?

(*) $w_0, w_1, w_2 \Rightarrow$ [Best decision Boundary]

(*) weights = importance of features

(*) Bias / intercept = position of Boundary

⇒

Scaling to n Dimension ⇒ Hyperplane

$w_1 x_1 + w_2 x_2 + w_0 = 0$

3D ⇒ $w_1 x_1 + w_2 x_2 + w_3 x_3 + w_0 = 0$

nD ⇒ $w_1 x_1 + w_2 x_2 + \cdots w_n x_n = 0$

hyperplane

100 features ⇒ 100 dim hyperplane

2D ⇒

$$w = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ \vdots \\ w_n \end{bmatrix}$$

Parallel & ⊥ar Lines

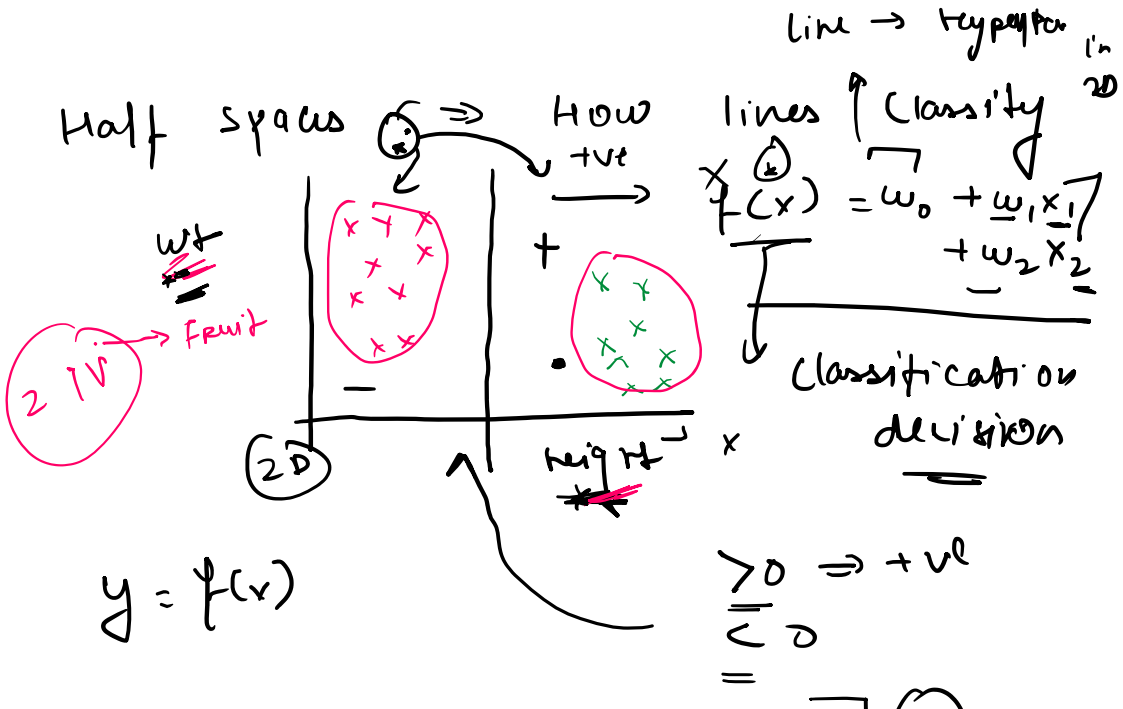
⇒ 2 lines parallel ⇒ same slope

⇒ 2 lines ⊥ar ⇒ $m_1 \times m_2 = -1$

⇒ $L_1 \Rightarrow y = \frac{3}{m_1} x + 4$ & $L_2 \Rightarrow 3y = 9x + 9$

|| or Lar

3 = 2 ⇒ ||



→ VECTOR

Physics

Magnitude

Direction

ML

(2,3)

(0,0)

[2, 3]

(0,0)

Point (2,3) → [2, 3]

ML → Data Point → vector

(1) Fruit ⇒ weight, shape, width → Data point → vector

100 Test Results → vector with 100 components

king — man — vector Machines → 0 1

vectors

Male ⇒ Similarity

Similarity: kind, Mas

⇒ VECTOR use cosθ

(2,3)

$\vec{x}$, $\vec{x}$

column vectors

$\vec{x} = [x_1, x_2]^T \Rightarrow$ R

$x \in R^2$ → x is a vector with 2 Real valued components

Real

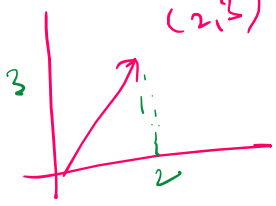
Physics

ML

R^3 → 3 Real comp

Magnitude → How long is the ARROW?

vector = [2, 3]



$\Rightarrow \sqrt{2^2+3^2} = \sqrt{13} = 3.6$

d dim^n vector → $\|\vec{x}\| = \sqrt{x_1^2 + x_2^2 + \cdots x_d^2}$

d – dim^n

→ position vector → (0, 0) origin

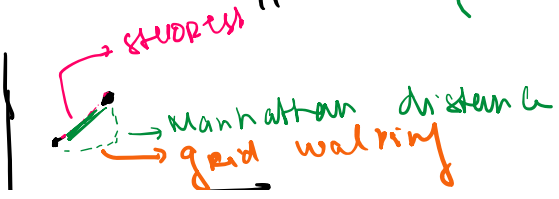
→

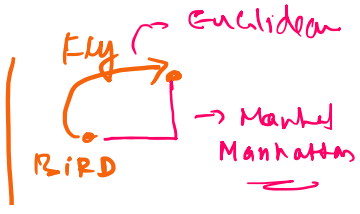
L2 NORM V/s L1 NORM

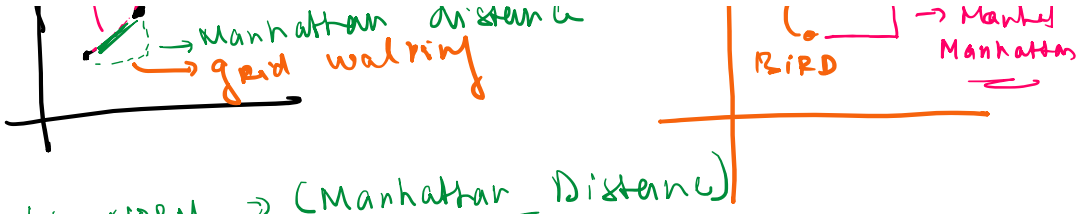
Magnitude → NORM

→ SHORTEST line

L2 NORM ⇒ (Euclidian) ⇒ $\|\vec{x}\| = \left(\sqrt{x_1^2 + x_2^2 + \cdots x_d^2}\right)^{1/2}$



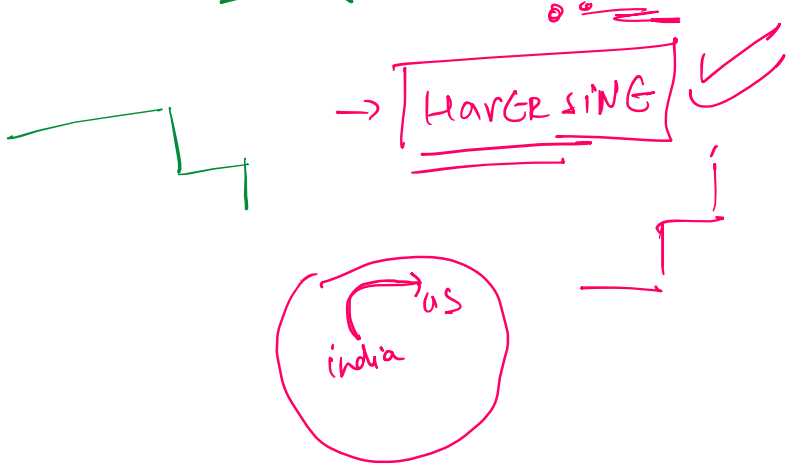


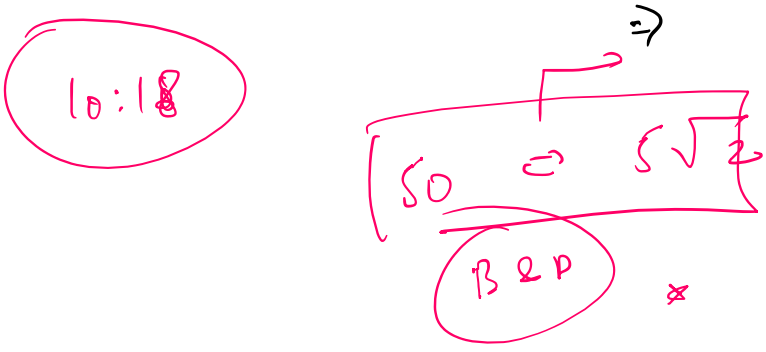


L1 NORM ⇒ (Manhattan Distance)

city

straight road





SHORTEST → EUCLIDEAN

CAR → Manhattan ⇒ city

(x_1, y_1) (x_2, y_2)

Manhattan = $(x_2 - x_1) + (y_2 - y_1)$

→ MATRIX MULTIPLICATION

FRWI

weight shape ---



→ VECTOR

Matrix MULTIPLICATION ⇒

Rule

$$[A]_{m \times n} \times [B]_{a \times b}$$

$n = a$

$$[C]_{m \times b}$$

(*) COLUMN VECTOR PROBLEM ⇒

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \text{ or } [x_1 \ x_2]^T$$

$$\vec{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \text{ or } [y_1 \ y_2]^T$$

$y = \begin{bmatrix} 0 \\ y_2 \end{bmatrix}$

$[x_1 \; x_2]^T_{2\times 1}$ $[y_1 \; y_2]^T_{2\times 1}$?

NO X

=

Simple Trx ⇒ Transpose one of them

$[x_1 \; x_2]^T = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

Row → Transpose

$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_{2\times 1}$ $[y_1 \; y_2]_{1\times 2}$ ⇒ vector embedding

∴ Yes

Transpose ⇒ $x_1 y_1 + x_2 y_2$ = Scalar value

DOT PRODUCT

$\vec{x}^T \cdot \vec{y}$ = DOT PRODUCT = $[x_1 y_1 + x_2 y_2]$ → Scalar value

10, 8 (25) ← scalar

$\vec{a} \times \vec{b}$

Cross Product ✗ ⇒ MU

Dot Product ⇒ ✓

Similarity

king [2, 3]

man [3, 5]

↑ Dot ⇒ ↑ Similarity

image → [0 0 0 0 10 15 20] ⇒ Matrix

↓

Dog

Loan Example ⇒

Loan approval ⇒

Features → Credit Score (x_1)

Years Employed (x_2)

Income (x_3)

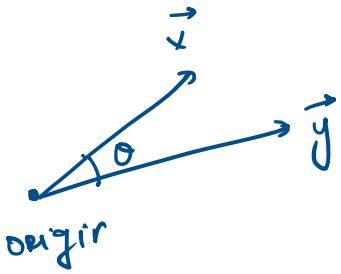
Trained weights ⇒ [10, 4, 700]

$pred^n = w^T x = 10x_1 + 4x_2 + 700x_3$

Credit score

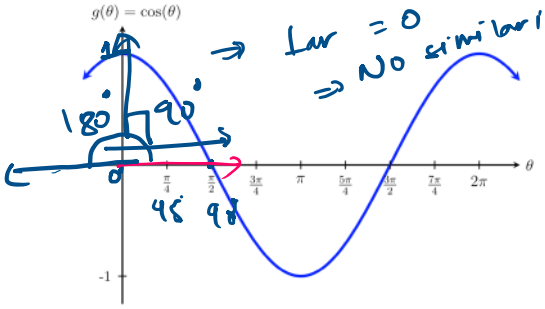
70x more influence

Angle b/w Vectors



$$\cos\theta = \frac{\text{dot product}}{\text{magnitude / product of norms}}$$

$$\cos\theta = \frac{\vec{x} \cdot \vec{y}}{\|\vec{x}\| \, \|\vec{y}\|}$$



$\Rightarrow$ Lw $= 0$
$\Rightarrow$ No similarity.

3 cases $\Rightarrow$

(1) $\cos\theta > 0$

$\Rightarrow \theta < 90°$

same direction

$\Rightarrow$ dot product $\rightarrow$ +ve

king $\rightarrow$ acute
Man

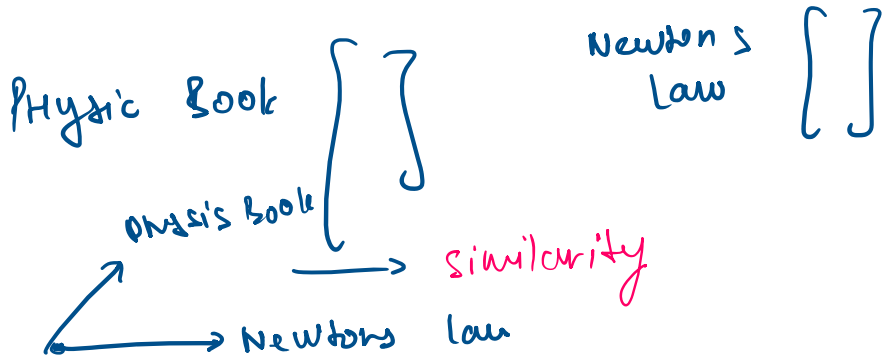
(2) $\cos\theta = 0 \Rightarrow \theta = 90°$

(3) $\cos\theta < 0 \Rightarrow$ angle $> 90°$

$\Rightarrow$ opposite direction

$\Rightarrow$ dot product $\rightarrow$ -ve

$\Rightarrow$ obtuse angle



some, No, opposite

$\vec{x} = [-2, 0]$ $\vec{y} = [2, 0]$

$\theta = ?$

$\vec{x} \cdot \vec{y} = (-2)(2) + 0 \times 0 = -4$

$||\vec{x}|| = 2$, $||\vec{y}|| = 2$

$\cos\theta = \frac{\vec{x} \cdot \vec{y}}{||x|| \, ||y||} = \frac{-4}{2 \times 2} = -1$

$\cos\theta = -1$

$\theta = 180^\circ$

Vectors a = [2, −1, 3] and b = [4, 8, −2]. Compute the dot product a·b. What does the SIGN of this result tell you about the angle between them?

From <https://www.scaler.com/instructor/meetings/i/linear-algebra-problem-solving-session/>

$\vec{a} = [2, -1, 3]$ $\vec{b} = [4, 8, -2]$

$\vec{a} \cdot \vec{b} = 2(4) + (-1)(8) + 3(-2)$

$= 8 - 8 - 6$

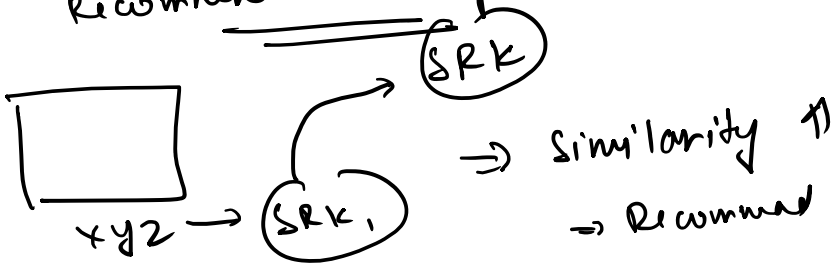
$= -6$

obtuse angle $180° \rightarrow -1$



→ document similarity

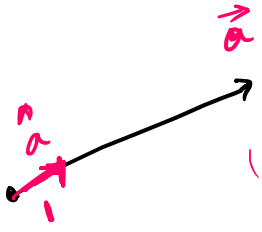
→ Recommendation engines



→ Unit vectors

⇒ magnitude exactly = 1

Captures only direction



$\hat{a}$ → unit vector

$\vec{w} = [3, 4] \Rightarrow \|w\| = \sqrt{3^2 + 4^2} = 5$

$\hat{w} = \frac{\vec{w}}{|\vec{w}|} = \left[\frac{3}{5}, \frac{4}{5}\right] = [0.6, 0.8]$ → $\sqrt{0.36 + 0.64}$

$\sqrt{0.81 + 0.64}$ $= \sqrt{1}$

$|\hat{w}|$ $\sqrt{0.81 + 0.64}$ $= \sqrt{1}$ $= 1$

unit vector

$\Rightarrow$ unit vector or not ?

— unit vector

$[1, 1, 1] \Rightarrow$ unit vector?

$\sqrt{1^2+1^2+1^2} = \sqrt{3} = 1.73$ ✗

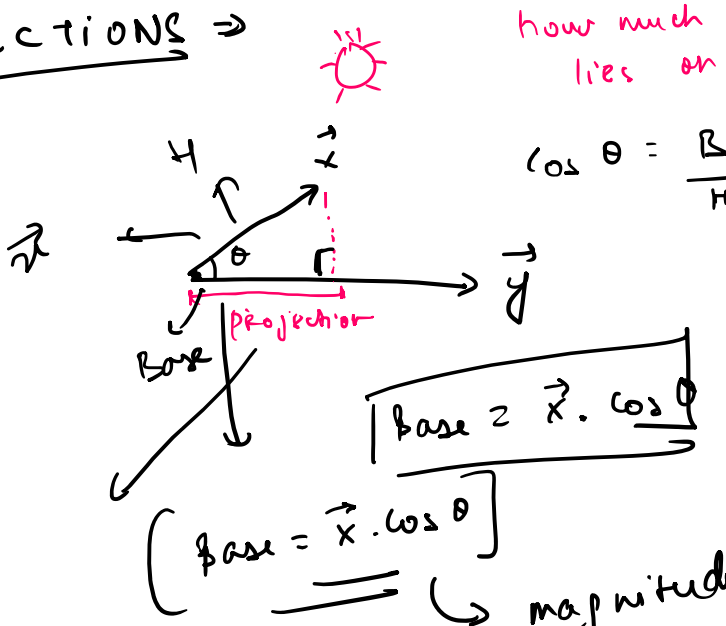
✗ unit vector

$\left[\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right]$ unit vector

0.577

⑧ PROJECTIONS ⇒

how much of $\vec{x}$ lies on $\vec{y}$



$\cos\theta = \frac{\text{Base}}{\text{Hypotenuse}}$

$\text{Base} = \vec{x} \cdot \cos\theta$

$\left(\text{Base} = \vec{x} \cdot \cos\theta\right)$ ⟹ magnitude

+ direction

Projection of $\vec{x}$ on $\vec{y}$ = $\vec{x} \cos\theta \cdot \hat{y}$ (unit vector)

Projection of x on y

$\hat{y} = \frac{\vec{y}}{|\vec{y}|}$

$\text{Proj} = \frac{|\vec{x}| \cos\theta \, \vec{y}}{|\vec{y}|}$

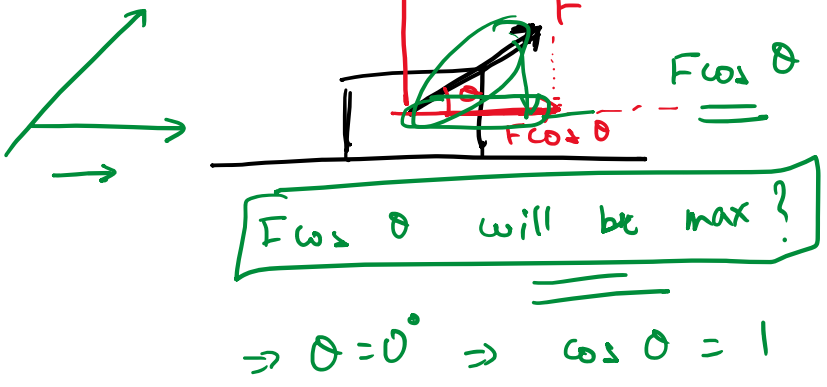
$\vec{x} \perp \vec{y}$

$\cos\theta = \frac{\vec{x} \cdot \vec{y}}{|\vec{x}| \, |\vec{y}|}$

$\text{Projection} = \frac{|\vec{x}| \, \vec{x} \cdot \vec{y}}{|\vec{x}| \, |\vec{y}| \, |\vec{y}|} \vec{y}$

$\left(\vec{x} \cdot \frac{\vec{y}}{|\vec{y}|}\right) \frac{\vec{y}}{|\vec{y}|}$

$= (\vec{x} \cdot \hat{y}) \hat{y}$ = Projection of x on y



$F = [F\cos\theta, F\sin\theta]$

$F\cos\theta$ will be max?

$\Rightarrow \theta = 0^\circ \Rightarrow \cos 0 = 1$